

Day 55 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography, Climate & Disasters (1)

READING MATERIAL

Covers India's recent extreme-weather and climate-disaster statistics, drawn from 2025-2026 climate reports — directly relevant to the syllabus's 'Natural calamity - Disaster management' and 'Climate change' clauses, updated with the latest available figures.

Q: How frequently did India face extreme weather events in 2025, per recent reporting?

A: India faced extreme weather on 331 of 334 days in 2025 — extreme weather events occurred on 99% of days between 1 January and 30 November 2025.

Q: What was the human and economic toll of these 2025 extreme weather events?

A: At least 4,419 deaths; around 17.4 million hectares of cropped land affected; at least 181,459 houses destroyed; and approximately 77,189 animals killed.

Q: Where does India rank on the Climate Risk Index 2026, and what are the associated losses?

A: 9th among the most climate-affected countries, with approximately 80,000 deaths and USD 170 billion in losses attributed to climate impacts.

Q: Where did India rank on the Climate Risk Index 2025 (covering the longer period 1993-2022)?

A: 6th most affected country globally over that period, with losses exceeding USD 180 billion.

Q: How has India's average temperature changed in recent decades, and what specific record was set in February 2025?

A: The Indian landmass was warmer by nearly 0.9°C during 2015-2024 compared to the 1901-1930 baseline. The India Meteorological Department (IMD) confirmed February 2025 as the hottest February on record.

Q: What change has been observed in Himalayan glacial lakes, and what risk does this raise?

A: Glacial lakes and other water bodies in the Himalayan region saw a 10.81% increase in area from 2011 to 2024 due to climate change — raising the risk of catastrophic Glacial Lake Outburst Floods (GLOFs).

Q: Approximately how many people in India have been internally displaced due to climate change, per the IDMC?

A: About 14 million people, per estimates from the Internal Displacement Monitoring Centre (IDMC).

MOCK TEST (WITH ANSWERS)

1. In 2025, India faced extreme weather events on approximately what percentage of days (Jan-Nov)?

- A. 50%
- B. 75%
- C. 90%
- D. 99% ✓**

Explanation: 99% of days. [medium, web-research]

2. Approximately how many deaths were attributed to extreme weather events in India in 2025?

- A. 1,200
- B. 2,800
- C. 4,419 ✓**
- D. 10,000

Explanation: 4,419 deaths. [hard, web-research]

3. On the Climate Risk Index 2026, India ranks among the most climate-affected countries at position:

- A. 3rd
- B. 6th
- C. 9th ✓**
- D. 15th

Explanation: 9th. [medium, web-research]

4. How much warmer was the Indian landmass during 2015-2024, compared to the 1901-1930 baseline?

- A. 0.3°C
- B. 0.5°C
- C. 0.9°C ✓**
- D. 1.5°C

Explanation: Nearly 0.9°C. [hard, web-research]

5. Which month did the IMD confirm as the hottest on record, in early 2025?

- A. January 2025
- B. February 2025 ✓**
- C. March 2025
- D. April 2025

Explanation: February 2025. [medium, web-research]

6. What was the increase in the area of Himalayan glacial lakes between 2011 and 2024?

- A. 2.5%
- B. 5.4%
- C. 10.81% ✓**
- D. 20%

Explanation: 10.81%. [hard, web-research]

7. An increase in Himalayan glacial lake area raises the risk of which specific disaster?

- A. Landslides only
- B. Earthquakes
- C. Glacial Lake Outburst Floods (GLOFs) ✓**
- D. Avalanches only

Explanation: Glacial Lake Outburst Floods (GLOFs). [medium, web-research]

8. Approximately how many people in India have been internally displaced due to climate change, per the IDMC?

- A. 2 million
- B. 6 million
- C. 14 million ✓**
- D. 25 million

Explanation: About 14 million. [hard, web-research]

Part B – Aptitude & Mental Ability: Compound Interest

READING MATERIAL

Compound Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC tests the standard CI formula, a useful SI-vs-CI shortcut, AND disguises CI as 'successive percentage change' word problems (depreciation, year-over-year growth) — recognize all three shapes.

Q: Formula: What is the Compound Interest formula, and the shortcut for the DIFFERENCE between SI and CI over 2 years?

A: Amount = $P(1+R/100)^T$, so CI = Amount – P. For exactly 2 years, there's a shortcut: $CI - SI = P \times (R/100)^2$. This avoids computing CI and SI separately when you only need their difference.

Q: Real exam (2022): Find the difference between SI and CI for 2 years on a principal of Rs.5,000 at 8% per annum.

A: Using the shortcut: difference = $P \times (R/100)^2 = 5000 \times (0.08)^2 = 5000 \times 0.0064 = \text{Rs.}32$. (Full check: SI = $5000 \times 8 \times 2/100 = 800$. CI = $5000 \times (1.08^2 - 1) = 5000 \times 0.1664 = 832$. Difference = $832 - 800 = 32$ ✓.)

Q: Real exam (2022): Find the compound interest on Rs.5,000 at 12% per annum for 2 years, compounded annually.

A: CI = $P \times ((1+R/100)^T - 1) = 5000 \times (1.12^2 - 1) = 5000 \times (1.2544 - 1) = 5000 \times 0.2544 = \text{Rs.}1,272$.

Q: Real exam (2025): A sum becomes Rs.13,380 after 3 years and Rs.20,070 after 6 years on compound interest. Find the original sum.

A: The amount after 6 years equals the amount after 3 years multiplied by the SAME growth factor again (since compounding is multiplicative): $(1+R/100)^3 = 20070/13380 = 1.5$. The sum itself: $P = \text{Amount-after-3-years} \div (1+R/100)^3 = 13380 \div 1.5 = \text{Rs.}8,920$.

Q: Real exam (2024): A motorcycle's value 2 years ago was Rs.80,000. It depreciates 4% in the first year, then increases 4% in the second year. Find the present value.

A: Apply each year's change as its own multiplier, in sequence — this is compound interest with a changing/mixed rate: $80,000 \times 0.96$ (first year, -4%) $\times 1.04$ (second year, +4%) = $80,000 \times 0.9984 = \text{Rs.}79,872$. Note the result is slightly LESS than the original Rs.80,000 — a decrease followed by an equal-percentage increase never fully cancels out, since the increase applies to a smaller base.

MOCK TEST (WITH ANSWERS)

1. Find the difference between SI and CI for 2 years on a principal of Rs.8,000 at 5% per annum.

- A. Rs.18
- B. Rs.20 ✓**
- C. Rs.22
- D. Rs.25

Explanation: Difference = $P \times (R/100)^2 = 8000 \times (0.05)^2 = 8000 \times 0.0025 = \text{Rs.}20$. [medium, authored]

2. Find the compound interest on Rs.10,000 at 10% per annum for 2 years, compounded annually.

- A. Rs.2,000
- B. Rs.2,100 ✓**
- C. Rs.2,200
- D. Rs.1,900

Explanation: CI = $10000 \times (1.10^2 - 1) = 10000 \times 0.21 = \text{Rs.}2,100$. [easy, authored]

3. A sum becomes Rs.16,000 after 2 years and Rs.19,360 after 4 years on compound interest. Find the original sum.

- A. Rs.12,500
- B. Rs.13,000
- C. Rs.13,223 ✓**
- D. Rs.14,000

Explanation: $(1+R/100)^2 = 19360/16000 = 1.21$. $P = 16000/1.21 \approx \text{Rs.}13,223$. [hard, authored]

4. A machine valued at Rs.50,000 depreciates by 10% in the first year and a further 10% in the second year. Find its value after 2 years.

- A. Rs.40,000
- B. Rs.40,500 ✓**
- C. Rs.41,000
- D. Rs.39,500

Explanation: $50,000 \times 0.90 \times 0.90 = 50,000 \times 0.81 = \text{Rs.}40,500$. [medium, authored]

5. Find the compound interest on Rs.20,000 at 15% per annum for 2 years, compounded annually.

A. Rs.6,000

B. Rs.6,450 ✓

C. Rs.6,200

D. Rs.6,500

Explanation: $CI = 20000 \times (1.15^2 - 1) = 20000 \times 0.3225 = \text{Rs.}6,450$. [medium, authored]

Part C – General English: Prose: "A Hunter Turned Naturalist" — Jim Corbett

READING MATERIAL

A biography of Jim Corbett — hunter, tracker, wildlife photographer, naturalist, and writer — who hunted man-eating tigers and leopards only at the Government's request to protect people, while genuinely admiring wildlife and refusing to hunt for sport.

Q: What was Jim Corbett famous for, and what was his dog's name?

A: Wildlife photography and hunting. His dog's name was Robin.

Q: What kind of animals did Jim Corbett hunt, and under what circumstances?

A: Man-eating tigers and leopards — and ONLY at the request of the Government, to help protect people, not for sport or pleasure.

Q: What was Corbett's attitude toward wildlife in general, despite being an expert hunter?

A: He admired wildlife and did not like to kill wild animals for pleasure or sport — but he did not hesitate to kill specifically man-eating tigers or leopards when needed to protect human lives.

Q: According to Corbett, when does a tiger typically become a man-eater, and what happens once a tiger has eaten human flesh?

A: A tiger typically becomes a man-eater when it gets OLD OR INJURED (and can no longer hunt its usual prey effectively). Once a tiger has eaten human flesh, it will NOT go back to eating other prey — it continues preying on humans.

Q: What was the Champawat Tiger, and what did Corbett's post-mortem reveal about it?

A: A Bengal tigress responsible for nearly 436 deaths across Nepal and the Kumaon area of India, shot by Corbett in 1907. The post-mortem revealed her upper and lower canine teeth on the right side were broken — an old injury from a previous gunshot by another hunter who had failed to track and kill her, which likely impaired her ability to hunt normal prey and turned her toward man-eating.

Q: What is Jim Corbett's lasting legacy regarding India's national parks?

A: He played a key role in establishing India's FIRST national park in the Kumaon Hills (originally named Hailey National Park, in the 1930s, in present-day Uttarakhand). It was renamed in 1957 as Jim Corbett National Park, in his honour. He died on 19 April 1955.

MOCK TEST (WITH ANSWERS)

1. Jim Corbett was famous for:

A. Architecture

B. Politics

C. Wildlife photography and hunting ✓

D. Sculpting

Explanation: Wildlife photography and hunting. [easy, web-research]

2. What kind of animals did Jim Corbett primarily hunt?

A. All wild animals indiscriminately

B. Only lions

C. Man-eating tigers and leopards ✓

D. Elephants

Explanation: Man-eating tigers and leopards. [medium, web-research]

3. What was the name of Jim Corbett's dog?

A. Tiger

B. Robin ✓

C. Bullet

D. Hunter

Explanation: Robin. [medium, web-research]

4. Jim Corbett hunted man-eating animals:

A. For sport

B. For fame

C. At the Government's request ✓

D. For survival

Explanation: At the Government's request. [medium, web-research]

5. According to Corbett, when does a tiger typically become a man-eater?

- A. When it gets angry
- B. When it's young

C. When it gets old or injured ✓

- D. When it lives alone

Explanation: When it gets old or injured. [medium, web-research]

6. What happens once a tiger has eaten human flesh, according to Corbett?

- A. It runs away permanently
- B. It dies soon after

C. It will not go back to eating other prey ✓

- D. It gets scared of humans

Explanation: It will not go back to eating other prey. [hard, web-research]

7. The Champawat Tiger, shot by Corbett in 1907, was responsible for approximately how many deaths?

- A. 100
- B. 250

C. 436 ✓

- D. 600

Explanation: 436 deaths. [hard, web-research]

8. What did the post-mortem reveal about the Champawat Tiger's old injury?

- A. A broken leg from a trap

B. Broken canine teeth from a previous gunshot ✓

- C. Blindness in one eye

- D. A poisoned wound

Explanation: Broken canine teeth from a previous gunshot. [hard, web-research]

9. Which national park, established with Jim Corbett's involvement, was later renamed in his honour (1957)?

A. Hailey National Park (now Jim Corbett National Park) ✓

- B. Gir National Park

- C. Kaziranga National Park

- D. Bandipur National Park

Explanation: Hailey National Park, renamed Jim Corbett National Park. [medium, web-research]